Lakshmi Manasa Maddi

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Education

University of California, San Diego [UCSD] **Sept. 2023 – Mar. 202**

Bachelor of Science in Data Science (Concentration: AI/ML), Minor: Business Analytics (Provost Honors) *San Diego, C*

- **Awards**: **1st Place** at **JPMorgan Chase Data for Good Hackathon** — made an educational analytics ML tool, **2nd Place** for **Data Science Student Society Annual Showcase** — built a time series + ML forecasting model, **Top 10** for **WIC Programming Competition (100+ teams)** — solved DS/Algorithm challenges in Python + Java, **1st Place** for **SDSU Big Data Hackathon (Geocomputational Thinker Award)** — built GIS-based ML app.

Experience

IBM (International Business Machines) **Jun. 2025 – Aug. 2025**

Artificial Intelligence Architect Intern — IBM National Market *Dallas, TX*

- Engineered **data pipelines** with Python and SQL using LangChain + Elasticsearch to unify **100K+ customer records**, achieving **95%+ accuracy** and enabling performance dashboards for onboarding and retention tracking.
- Developed a **Generative AI RAG agent** within Watsonx Orchestrate that automated insights across Salesforce and Slack, cutting **workflow time by 70%** and improving Customer Success team productivity by **15+ hours per week**.
- Performed **churn and renewal analytics** through hypothesis testing and time series modeling, uncovering a **12% higher churn likelihood** for delayed onboardings — insights adopted into a new product adoption strategy.

Google X Basta **Jun. 2025 – Aug. 2025**

Google SWE Program Fellow (Basta G-SWEP) *Remote*

- Built and optimized **Python-based ETL scripts** and **SQL queries** to clean and analyze project datasets, improving data accessibility and efficiency for team experiments; collaborated with Google engineers to review code structure and strengthen **software engineering best practices**.

Halıcıoğlu Data Science Institute (HDSI) Tilos AI Lab **Jan. 2025 – Jun. 2025**

Undergraduate Data Science Research Fellow *San Diego, CA*

- Built **gesture recognition models** on wearable sensor data using **KNN** and **decision trees** in **scikit-learn**, reaching **91% accuracy** and revealing motion patterns that improved the lab's real-time human–AI interaction studies.
- Created reproducible workflows with **Python** for preprocessing and **SQL** for data extraction, fine-tuned **transformer models** for text prediction, and automated **Tableau dashboards** that cut reporting time by **38%**.

Wild Genomics **Jul. 2024 – Sep. 2024**

Data Science Intern *San Diego, CA*

- Optimized genomic data workflows on **AWS Spark**, scaling processing of **50K+ DNA samples** and reducing runtime by **30%**, which accelerated biodiversity analysis.
- Applied **regression, clustering, and anomaly detection** to classify species and flag sequencing errors, improving model accuracy by **18%** and validating results with **SQL checks**.
- Built interactive **dashboards** in **Power BI**, **Looker**, and **AWS QuickSight** to track species diversity and population health, reducing manual reporting by **50%** and enabling scientists to make conservation decisions faster.

Projects

Modeling Economic Trajectories Analysis | *Python, XGBoost, Time Series, SQL, Tableau* **Jun. 2025**

- Built **hybrid ML and time series models** (**XGBoost**, **SARIMA**) on global economic data, achieving $\mathbf{R^2 = 0.96}$ and outperforming baseline forecasts by over **150%**.
- Identified key growth drivers using **lag features, rolling statistics, and SHAP analysis**; developed reproducible workflows in **Python** and **SQL**, and built interactive **Tableau dashboards** to communicate insights. Project Link

National Power Outage Duration Predictor | *Python, SQL, MATLAB, Tableau* **Apr. 2025**

- Analyzed **1,500+ U.S. outage events** with **regression** and **anomaly detection models** in **Python** and **MATLAB**, improving prediction accuracy by **12%**.
- Streamlined data processing with **SQL** and **Python**, and deployed automated **Tableau dashboards** that reduced reporting time by **30%** and improved decision-making speed. Project Link

Technical Skills

Programming & Data Systems: Python (Pandas, NumPy, Scikit-learn, LangChain), R, SQL, PostgreSQL, Java
Machine Learning & AI: Regression, Classification, Forecasting, Clustering, Decision Trees, Large Language Models (LLMs/SLMs), Agents, Deep Learning, Statistical Modeling, MATLAB, Excel, Hadoop, Spark
Analytics & Experimentation: A/B Testing, Hypothesis Testing, Experiment Design, Causal Inference, Product & Service Telemetry, Data Exploration, Interpretation & Validation, Monitoring & Iteration, Econometrics
Statistics & Operations Research: Probability, Sampling, Inference, Optimization, Simulation, Time Series Analysis
Visualization & Reporting: Tableau, Power BI, Looker, Matplotlib, Automated Dashboards, Experiment Tracking, Metric Design, AWS, Data Storytelling & Communication